Further Interpolation Methods for Doppler Scale Estimation in Underwater Acoustic CP-OFDM Systems

Muhammad Muzzammil
College of Underwater Acoustic Engineering,
Harbin Engineering University
Harbin 150001, China
e-mail: muzzammilm@hrbeu.edu.cn

Hanbo Jia
College of Underwater Acoustic Engineering,
Harbin Engineering University
Harbin 150001, China
e-mail: 496899041@hrbeu.edu.cn

Lei Wan
College of Underwater Acoustic Engineering,
Harbin Engineering University
Harbin 150001, China
e-mail: leiwan@hrbeu.edu.cn

Gang Qiao
College of Underwater Acoustic Engineering,
Harbin Engineering University
Harbin 150001, China
e-mail: qiaogang@hrbeu.edu.cn

***Abstract*—Accurate Doppler scale estimation and compensation in the challenging underwater acoustic (UWA) communication channels are the essential requirement for the performance of the orthogonal frequency division multiplexing (OFDM) systems. In UWA OFDM systems, increasing the correlation accuracy between the received cyclic prefix (CP) and OFDM symbol for Doppler scale estimation depends on the sampling rate of the signal. Increased sampling rate results in an increased accuracy at the cost of an increased computational complexity. Therefore, in this paper, we adopted a less computational complex auto-correlation approach, where we first derived the closed form expression for auto-correlation based Doppler scale estimation in CP-OFDM signal, and then proposed three different fine Doppler scale estimation methods. From the simulations, it is observed that in single path channel, methods 2 and 3 perform better in terms of RMSE and BER, while method 1 shows better performance in multipath channels.**

Keywords-Underwater acoustic (UWA) communication; Doppler scale estimation; orthogonal frequency division multiplexing (OFDM); cyclic prefix (CP)

I. INTRODUCTION

Orthogonal frequency division multiplexing (OFDM) is being widely used for underwater acoustic communications (UWAC) in the last decade and becomes a suitable choice in short range and high data rates over channels with long delay spread [1]–[3]. The underwater acoustic (UWA) channel is very challenging due to the low propagation speed of sound in water (i.e., 1500*m/s*) and small carrier frequency to bandwidth ratio, which leads to large multipath and substantial Doppler effect in UWAC. Therefore, Doppler effect must be dealt properly when applying OFDM in UWA channels.

In the literature, various methods such as Doppler insensitive waveform, Doppler sensitive waveform and least square (LS) methods have been offered to estimate the common Doppler scale factor [2], [4]–[8]. Doppler insensitive waveforms such as linear-frequency modulation waveform and hyperbolic-frequency modulation waveform are adopted as the preamble and postamble of the signal structure to estimate the common Doppler scale factor [2], [4], [5]. Doppler scale factor is estimated by calculating the duration between preamble and postamble through a single branch matched filter and compare it with the original duration [9]. The drawback of using Doppler insensitive waveform method is that it introduces signal overhead and requires the whole data to be processed first in order to estimate the parameters, which may not be practical for online receiving process. Doppler sensitive waveform method which includes m-sequence coded waveforms and Costas waveforms can be employed as a preamble to estimate the Doppler scale factor. In the receiver side of this method, a bank of cross-correlators are used and the correlator yielding the largest peak opted as the estimate of the Doppler scale factor [10]. This method also have drawbacks of additional signal overhead and more computational complexity. Therefore, there is a need of method to estimate the Doppler scale factor which uses minimum signal overhead as well as less computational complexity.

The auto-correlation based Doppler scale factor estimation has been extensively adopted in the literature, which utilizes cyclic prefix (CP) in CP-OFDM systems without introducing extra signals overhead, and it is also a less computational complex approach [11]–[15]. The estimation accuracy of the Doppler scale estimation depends on the sampling rate of the received signal when directly applying auto-correlation. The received signal may requires to be oversampled to improve the estimation accuracy at the cost of more computational complexity. An interpolation technique has been used to improve the estimation accuracy without increasing the computational complexity [11], which is copied from the frequency estimation [16], [17]. In this

978-1-7281-5102-1/19/$31.00 ©2019 IEEE

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on April 19,2026 at 09:17:59 UTC from IEEE Xplore. Restrictions apply.

CP	X	X

Figure 1. CP-OFDM signal structure

paper, we focuses on a multipath UWA channel having a common Doppler scale in all paths and derived a closed form expression of expected auto-correlation for the received CP-OFDM signal. Further, Doppler scale factor is estimated using interpolation methods which are derived from the expected auto-correlation closed form expression. In addition, simulations are carried out to validate the performances of the derived Doppler scale estimation methods under different downsampling factors.

The rest of the paper is organized as follows: Section II describes the system model and the proposed Doppler scale factor estimation methods are given in Section III. In Section IV, the simulations results are presented and Section V concludes the paper.

II. SYSTEM MODEL

A. CP-OFDM System and Multipath Channel Model

The CP-OFDM signal structure with two identical OFDM symbols is considered, as shown in Fig. 1. Let $s[k]$ to be the phase shift keying (PSK) / quadrature amplitude modulation (QAM) modulated symbol on the k th subcarrier, K is the total number of subcarrriers, $f_k = f_c + \frac{k}{T}$ is the frequency at the k th subcarrier, where f_c and T are the carrier frequency and OFDM symbol duration respectively. Then, continuous time CP-OFDM signal in passband, $x(t)$ can be written as

$$x(t) = \mathrm{Re}\left\{ \sum_{k=-\frac{K}{2}}^{\frac{K}{2}-1} s[k] e^{j2\pi f_k t} q(t) \right\}, \tag{1}$$

where $q(t)$ is the rectangular pulse shaping window, which depends on the CP-OFDM signal structure. According to our considered CP-OFDM signal structure, $q(t)$ will be

$$q(t) = \begin{cases} 1, & t \in [-T_{cp}, 2T]; \\ 0, & \text{elsewhere} \end{cases}$$

where, T_{cp} is the CP duration.

The channel impulse response of a multipath channel with L paths can be written as

$$h(t) = \sum_{l=0}^{L-1} \beta_l \delta[(1+\alpha)t - \tau_l],$$

where β_l, τ_l represent the amplitude and time delay of the l th path respectively, and α denotes the common Doppler scaling factor shared by all paths.

After the channel, the received passband signal is downshifted and low pass filter is applied, which leads to a baseband signal and can be written as

$$\begin{aligned} y(t) = & \sum_{l=0}^{L-1} \beta_l^c e^{j2\pi f_c \alpha t} \sum_{k=-\frac{K}{2}}^{\frac{K}{2}-1} s[k] e^{-j2\pi \frac{k}{T}\tau_l} e^{j2\pi \frac{k}{T}(1+\alpha)t} \\ & \times q[(1+\alpha)t - \tau_l] + w(t), \end{aligned} \tag{2}$$

where $\beta_l^c = \beta_l e^{-j2\pi f_c \tau_l}$ denotes the baseband equivalent complex path amplitude, and $w(t)$ represents the baseband noise.

B. Auto-Correlation Based Doppler Scale Estimation

The received baseband signal in (2) exhibits a repetition structure with period of $T/(1+\alpha)$, which can be expressed as [11]

$$y(t) = e^{-j2\pi \frac{\alpha}{1+\alpha} f_c T} y(t + \frac{T}{1+\alpha}), \tag{3}$$

where $t \in [-\frac{T_{cp} - \tau_{\max}}{1+\alpha}, \frac{T}{1+\alpha}]$, $\tau_{\max}$ represents the maximum delay spread. Therefore, the Doppler scale factor estimation based on the auto-correlation can be written as

$$\begin{aligned} \alpha &= \arg\max_{\alpha} |\Phi(\alpha)| \\ &= \arg\max_{\alpha} \left| \int_0^{T/(1+\alpha)} y(t) y^*(t + \frac{T}{1+\alpha}) \right|. \end{aligned} \tag{4}$$

The sampled digital signal can be obtained from (2) by assuming the signal is sampled at $f_s = \lambda B = \frac{\lambda K}{T}$, where B denotes the bandwidth of the OFDM signal, and λ represents the oversampling factor. The sampled signal can be expressed as

$$\begin{aligned} y[n] = & \sum_{l=0}^{L-1} \beta_l^c e^{j2\pi \varepsilon_\alpha \frac{n}{\lambda K}} \sum_{k=-\frac{K}{2}}^{\frac{K}{2}-1} s[k] e^{-j2\pi \frac{k}{T}\tau_l} e^{j2\pi \frac{k}{\lambda K}(1+\alpha)n} \\ & + w(n), \end{aligned} \tag{5}$$

where $\varepsilon_\alpha = f_c \alpha T$ is the normalized Doppler shift.

Then the expected auto-correlation output $\overline{\Phi}_D^\alpha[\Delta N]$ for Doppler scale estimation can be deduced from (5) and can be expressed as

$$\begin{aligned} & \overline{\Phi}_{\mathrm{D}}^{\alpha}[\Delta N] \\ & = E\{y[n] y^*[n+\Delta N]\} \\ & = E\{\sum_l \beta_l^c e^{j2\pi \varepsilon_\alpha \frac{n}{\lambda K}} \sum_{k=-\frac{K}{2}}^{\frac{K}{2}-1} s[k] e^{-j2\pi \frac{k}{T}\tau_l} e^{j2\pi \frac{k}{\lambda K}(1+\alpha)n} \\ & \sum_m (\beta_m^c)^* e^{-j2\varepsilon_\alpha \frac{n+\Delta N}{\lambda K}} \\ & \sum_{d=-\frac{K}{2}}^{\frac{K}{2}-1} s^*[d] e^{j2\pi \frac{d}{T}\tau_m} e^{-j2\pi \frac{d}{\lambda K}(1+\alpha)(n+\Delta N)} \} + E\{\Omega(\Delta N)\}, \end{aligned} \tag{6}$$

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on April 19,2026 at 09:17:59 UTC from IEEE Xplore. Restrictions apply.

where $\Omega(\Delta N)$ represents all the terms related with noise $w(n)$. The range of the considered CP-OFDM signal structure for calculating the expected value of auto-correlation could be taken as $n \in [-\frac{T_{cp}-\tau_{\max}}{1+\alpha} f_s, \frac{T}{1+\alpha} f_s]$.

In order to derive a closed form expression for the expected auto-correlation output, we use the assumptions of: The transmit symbols $s[k]$ are independent identically distributed (i.i.d.); The baseband equivalent complex path amplitudes β_l^c are independent distributed, with mean value $E\{\beta_l^c\} \approx 0$; The noise $w(n)$ is white, independent from the signal, and with zero mean. Considering these assumptions, the expected auto-correlation output in (6) can be re-written as

$$\begin{aligned}
&\overline{\Phi}_{\mathrm{D}}^{\alpha}[\Delta N] \\
&\approx \sum_l \delta_{\beta_l}^2 e^{-j2\pi\varepsilon_\alpha \frac{\Delta N}{\lambda K}} \sum_{k=-\frac{K}{2}}^{\frac{K}{2}-1} \delta_s^2 e^{-j2\pi \frac{k(1+\alpha)\Delta N}{\lambda K}} \qquad (7) \\
&= \delta_s^2 \sum_l \delta_{\beta_l}^2 e^{-j2\pi\varepsilon_\alpha \frac{\Delta N}{\lambda K}} e^{j\pi \frac{(1+\alpha)\Delta N}{\lambda K}} \frac{\sin \pi \frac{(1+\alpha)\Delta N}{\lambda}}{\sin \pi \frac{(1+\alpha)\Delta N}{\lambda K}}.
\end{aligned}$$

The absolute value of $|\Phi_\alpha[\Delta N]|$ can be expressed as

$$\begin{aligned}
\left|\overline{\Phi}_{\mathrm{D}}^{\alpha}[\Delta N]\right| &= \delta_s^2 \sum_l \delta_{\beta_l}^2 \left| \frac{\sin \pi \frac{(1+\alpha)\Delta N}{\lambda}}{\sin \pi \frac{(1+\alpha)\Delta N}{\lambda K}} \right| \qquad (8) \\
&= C\left|\Psi_\alpha[\Delta N]\right|,
\end{aligned}$$

where $\Psi_\alpha[\Delta N] = \frac{\sin \pi \frac{(1+\alpha)\Delta N}{\lambda}}{\sin \pi \frac{(1+\alpha)\Delta N}{\lambda K}}$, and $C = \delta_s^2 \sum_l \delta_{\beta_l}^2$ is a constant not related with α. Therefore, the accurate Doppler scale factor estimation can be found from the $\Psi_\alpha[\Delta N]$ function of the expected auto-correlation output.

III. Proposed Doppler Scale Estimation Methods

The proposed Doppler scale factor estimation methods uses three neighboring samples that are $\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0]$, $\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0 - 1]$ and $\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0 + 1]$, where $\Delta N_0 = \arg\max_{\Delta N} \left|\Phi_D^{\alpha}[\Delta N]\right|$, near the peak of auto-correlation outputs, similar to single frequency tone estimation [18]. Then, interpolation is carried out to find the estimate of the Doppler scale factor α. Next, we presents three different methods for Doppler scale factor estimation and are follows:

A. Method 1: Quadratic Function Fitting

In this method, the $|\Psi_\alpha[\Delta N]|$ function is approximated as a quadratic function to estimate the Doppler scale factor α. With this approximation, (8) can be written as

$$\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N]\right| = C \left| \frac{\sin \pi \frac{(1+\alpha)\Delta N}{\lambda}}{\sin \pi \frac{(1+\alpha)\Delta N}{\lambda K}} \right| \approx a\Delta N^2 + b\Delta N + c, \qquad (9)$$

where a, b, and c represents the parameters of the quadratic function. Then, the peak value of $\left|\Phi_D^{\alpha}[\Delta N_0]\right|$ can be expressed as [16]

$$\Delta N' = \Delta N_0 + \frac{0.5\left(\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0 + 1]\right| - \left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0 - 1]\right|\right)}{2\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0]\right| - \left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0 - 1]\right| - \left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0 + 1]\right|}. \qquad (10)$$

Equation (10) is the solution adopted for Doppler scaling factor estimation in [11]. Finally, the Doppler scale factor estimation using quadratic function fitting method can be expressed as

$$\alpha_{\text{method1}} = \frac{\lambda K}{\Delta N'} - 1. \qquad (11)$$

As this method is not derived based on the specific value of oversampling factor λ, therefore, it can be applied to any range of oversampling factors. The estimation accuracy is higher when λ is larger, i.e., the three samples of the interpolation should be close to each other.

The λ plays a significant role in deciding whether the $|\Psi_\alpha[\cdot]|$ function is inside the main lobe or not. For $\lambda > 1$, it is sure that $|\Psi_\alpha[\cdot]|$ function is inside the main lobe at ΔN_0, $\Delta N_0 - 1$ and $\Delta N_0 + 1$, because the main lobe width is $\frac{2\lambda}{1+\alpha}$ of $|\Psi_\alpha[\cdot]|$ function and typically the value of α is very small, i.e., $|\alpha| < 10^{-2}$. The rest of the two Doppler scale factor estimation methods are based on $\lambda > 1$.

B. Method 2

The $|\Psi_\alpha[\cdot]|$ function have the same sign near peak values for the samples ΔN_0, $\Delta N_0 - 1$, $\Delta N_0 + 1$, which means that three samples are negative for even K and positive for odd K, when $\lambda > 1$, With an assumptions that K is even where all three samples will be negative and α is usually very small, i.e., $|\alpha| < 10^{-2}$, the absolute values of the three samples can be written as

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on April 19,2026 at 09:17:59 UTC from IEEE Xplore. Restrictions apply.

$$\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0-1]\right| = -C\frac{\sin(\eta-\pi\frac{1+\alpha}{\lambda})}{\sin(\frac{\eta}{K}-\pi\frac{1+\alpha}{K\lambda})} \approx -C\frac{\sin(\eta-\gamma)}{\sin(\frac{\eta}{K}-\frac{\gamma}{K})},$$
$$\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0]\right| = -C\frac{\sin\eta}{\sin\frac{\eta}{K}}, \tag{12}$$
$$\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0+1]\right| = -C\frac{\sin(\eta+\pi\frac{1+\alpha}{\lambda})}{\sin(\frac{\eta}{K}+\pi\frac{1+\alpha}{K\lambda})} \approx -C\frac{\sin(\eta+\gamma)}{\sin(\frac{\eta}{K}+\frac{\gamma}{K})},$$

where $\eta = \pi\frac{(1+\alpha)\Delta N_0}{\lambda}$ and $\gamma = \frac{\pi}{\lambda}$.

Using Taylor series expansion for the denominators and ignoring high order terms in (12), it can be re-written as

$$\sin\frac{\eta}{K} \approx \pi - \frac{\eta}{K},$$
$$\sin(\frac{\eta}{K}-\frac{\gamma}{K}) \approx \pi - \frac{\eta}{K} + \frac{\gamma}{K}, \tag{13}$$
$$\sin(\frac{\eta}{K}+\frac{\gamma}{K}) \approx \pi - \frac{\eta}{K} - \frac{\gamma}{K},$$

By substituting the approximations in (13) into (12), and after some manipulations, we can get the following

$$\frac{\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0+1]\right|}{\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0]\right|}\frac{\pi-\frac{\eta}{K}-\frac{\gamma}{K}}{\pi-\frac{\eta}{K}} = \cos\gamma + \cot\eta\sin\gamma,$$
$$\frac{\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0-1]\right|}{\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0]\right|}\frac{\pi-\frac{\eta}{K}+\frac{\gamma}{K}}{\pi-\frac{\eta}{K}} = \cos\gamma - \cot\eta\sin\gamma. \tag{14}$$

We can obtain expression for variable η by adding the equations in (14), that is

$$\eta = \frac{\mu_1}{\mu_2},$$

where

$$\mu_1 = (K\pi-\gamma)\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0+1]\right| + (K\pi+\gamma)\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0-1]\right| - 2K\pi\cos\gamma\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0]\right|,$$

and

$$\mu_2 = \left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0+1]\right| + \left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0-1]\right| - 2\cos\gamma\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0]\right|.$$

Finally, the Doppler scale factor estimation can be written as

$$\alpha_{\text{method2}} = \frac{\lambda\eta}{\pi\Delta N_0} - 1 = \frac{\lambda\mu_1}{\pi\Delta N_0\mu_2} - 1. \tag{15}$$

C. Method 3

Based on (12), another analytical closed form expression can be obtained for Doppler scale estimation without using Taylor series expansion approximations.

Equations in (12) can also be written as

$$C\sin(\eta-\gamma) + C\sin(\eta+\gamma) = -\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0-1]\right|\sin(\frac{\eta}{K}-\frac{\gamma}{K}) - \left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0+1]\right|\sin(\frac{\eta}{K}+\frac{\gamma}{K}). \tag{16}$$

On the other hand,

$$C\sin(\eta-\gamma) + C\sin(\eta+\gamma) = 2C\sin\eta\cos\gamma = -2\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0]\right|\sin\frac{\eta}{K}\cos\lambda. \tag{17}$$

By combining (16) and (17), we have

$$2\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0]\right|\sin\frac{\eta}{K}\cos\gamma = \left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0-1]\right|\sin(\frac{\eta}{K}-\frac{\gamma}{K}) + \left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0+1]\right|\sin(\frac{\eta}{K}+\frac{\gamma}{K}). \tag{18}$$

Let $\Gamma_1 = \frac{\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0-1]\right|}{\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0]\right|}$ and $\Gamma_2 = \frac{\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0+1]\right|}{\left|\Phi_{\mathrm{D}}^{\alpha}[\Delta N_0]\right|}$, then η can be derived as the following

$$\eta = K\times\operatorname{atan}\{\frac{(\Gamma_2-\Gamma_1)\sin\frac{\gamma}{K}}{2\cos\gamma-(\Gamma_1+\Gamma_2)\cos\frac{\gamma}{K}}\}. \tag{19}$$

Substituting the value of η in (19) and after some manipulations, we get the final solution for method 3 and can be expressed as

$$\alpha_{\text{method3}} = \frac{\lambda K}{\pi\Delta N_0}\operatorname{atan}\left\{\frac{(\Gamma_2-\Gamma_1)\sin\frac{\gamma}{K}}{2\cos\gamma-(\Gamma_1+\Gamma_2)\cos\frac{\gamma}{K}}\right\}\text{-1}. \tag{20}$$

IV. Simulation Results and Analysis

Next, we provide several simulations to assess the performance of the proposed Doppler scale factor estimation methods, with the considered performance metrics of bit error rate (BER) and root mean squared error (RMSE) of Doppler speed as shown in (21). The direct estimation in the simulations means, the Doppler scale factor is estimated without using interpolation.

$$\text{RMSE}=\sqrt{E\left[\left|(\alpha-\alpha)\times c_s\right|^2\right]}. \tag{21}$$

where c_s is the sound speed in water.

The simulations setup of CP-OFDM system is given in Table. I and the channel settings are: the paths amplitudes and inter-arrival time are Rayleigh distributed and exponential distributed with mean 1 ms respectively, Doppler scale factors follow uniform distribution in the region of $[-1\times10^{-3}, 1\times10^{-3}]$.

In Fig. 2, the RMSE simulation results corresponding to $\lambda = 2,3,4$ in single path channel are presented. From Fig. 2,

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on April 19,2026 at 09:17:59 UTC from IEEE Xplore. Restrictions apply.

it can be seen that method 1, 2 and 3 have similar performance, however, in high SNR region method 2 and 3

TABLE I. SIMULATIONS SETUP OF CP-OFDM SYSTEM

Parameter(s)	Distribution/Value
Total number of subcarriers	1024
Bandwidth	6000 Hz
Symbol duration	170.7 ms
CP duration	50 ms

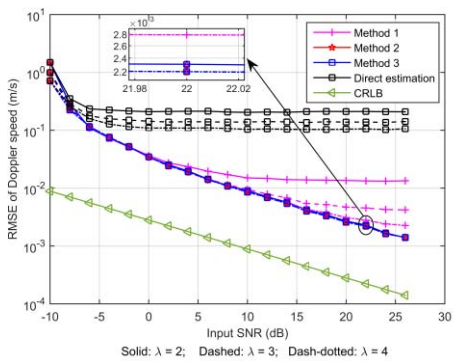

Figure 2. RMSE Performance: when $\lambda = 2,3,4$ and number of paths = 1

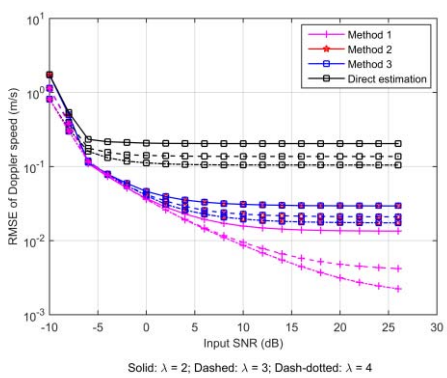

Figure 3. RMSE Performance: when $\lambda = 2,3,4$ and number of paths = 20

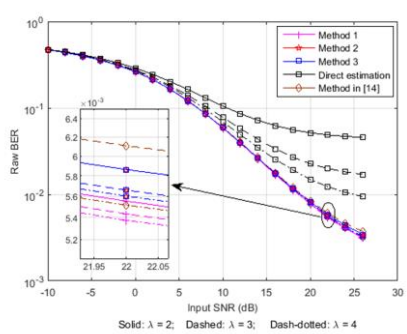

Figure 4. Raw BER Performance: when $\lambda = 2,3,4$ and number of path channel = 20

outperforms method 1. Overall, in single path channel, the proposed methods 2 and 3 shows similar and better performance amongst the considered methods, while direct estimation method shows worst performance. In Fig. 3, multipath channels (paths = 20) are considered and simulations results for RMSE corresponding to $\lambda = 2,3,4$ are provided. From Fig. 3, it can be observed that method 1 outperforms while direct estimation method shows the poorest performances amongst all the considered estimation methods. Further, it can be seen that method 2 and 3 shows the similar degraded performances, which is due to the model mismatch. From both Fig. 2, 3, it can be analyzed that, method 1 has the similar performance in both single and multipath channels settings, because they are not derived from the specific auto-correlation output functions.

Fig. 4 presents the raw BER performances of the considered Doppler scale factors interpolation methods in multipath channels (paths = 20). It can be seen from Fig. 4, that almost similar performances are shown by all the considered methods except the direct estimation method. The reason for such results are that the estimation methods 1, 2 and 3 for $\lambda \geq 2$ calculates the Doppler scale factor quite accurately.

V. CONCLUSION

In this paper, a closed form expression based on autocorrelation is derived for Doppler scale factor estimation in UWA CP-OFDM systems. The derived auto-correlation expression is further used and three different methods are proposed for Doppler scale factor estimations when the oversampling factor is greater than 1. Simulations are carried out to assess the RMSE and BER performances of the proposed methods for different oversampling factors in single and multipath channels environment. It is concluded that, the RMSE performance of method 2 and 3 is better in single path channel, while method 1 outperforms in multipath channels.

REFERENCES

[1] M. Stojanovic, "Low complexity OFDM detector for underwater acoustic channels," in *OCEANS* 2006, Sep. 2006, pp. 1–6.

[2] B. Li, S. Zhou, M. Stojanovic, L. Freitag, and P. Willett, "Multicarrier communication over underwater acoustic channels with nonuniform Doppler shifts," *IEEE Journal of Oceanic Engineering*, vol. 33, no. 2, pp. 198–209, April 2008.

[3] S. Zhou and Z. Wang, OFDM *for underwater acoustic communications*. John Wiley & Sons, 2014.

[4] S. A. Kramer, "Doppler and acceleration tolerances of high-gain, wideband linear FM correlation sonars," *Proceedings of the IEEE*, vol. 55, no. 5, pp. 627–636, May 1967.

[5] J. J. Kroszczynski, "Pulse compression by means of linear-period modulation," *Proceedings of the IEEE*, vol. 57, no. 7, pp. 1260–1266, July 1969.

[6] J. G. Proakis and M. Salehi, *Digital communications*. McGraw-hill New York, 2001, vol. 4.

[7] Q. Lu, Y. Huang, Z. Wang, and S. Zhou, "Characterization and receiver design for underwater acoustic channels with large Doppler spread," in *OCEANS 2015 - MTS/IEEE Washington*, Oct 2015, pp. 1–6.

[8] W. Pan, P. Liu, F. Chen, F. Ji, and J. Feng, "Doppler-shift estimation of flat underwater channel using data-aided least-square approach,"

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on April 19,2026 at 09:17:59 UTC from IEEE Xplore. Restrictions apply.

International Journal of Naval Architecture and Ocean Engineering, vol. 7, no. 2, pp. 426–434, 2015.

[9] B. S. Sharif, J. Neasham, O. R. Hinton, and A. E. Adams, "A computationally efficient Doppler compensation system for underwater acoustic communications," *IEEE Journal of Oceanic Engineering*, vol. 25, no. 1, pp. 52–61, Jan 2000.

[10] Q. Jin, K. M. Wong, and Z. Q. Luo, "The estimation of time delay and Doppler stretch of wideband signals," *IEEE Transactions on Signal Processing*, vol. 43, no. 4, pp. 904–916, April 1995.

[11] S. F. Mason, C. R. Berger, S. Zhou, and P. Willett, "Detection, synchronization, and Doppler scale estimation with multicarrier waveforms in underwater acoustic communication," *IEEE Journal on Selected Areas in Communications*, vol. 26, no. 9, pp. 1638–1649, December 2008.

[12] B.-C. Kim and I.-T. Lu, "Parameter study of OFDM underwater communications system," in *OCEANS 2000 MTS/IEEE Conference and Exhibition. Conference Proceedings (Cat. No.00CH37158)*, vol. 2, Sep. 2000, pp. 1251–1255.

[13] L. Ma, G. Qiao, and S. Liu, "A combined Doppler scale estimation scheme for underwater acoustic OFDM system," *Journal of Computational Acoustics*, vol. 23, no. 04, p. 1540004, 2015.

[14] Z. Chen, Y. R. Zheng, J. Wang, and J. Song, "Synchronization and Doppler scale estimation with dual PN padding TDS-OFDM for underwater acoustic communication," in *2013 OCEANS - San Diego*, Sep. 2013, pp. 1–4.

[15] N. Parrish, S. Roy, and P. Arabshahi, "Symbol by symbol Doppler rate estimation for highly mobile underwater OFDM," in *Proceedings of the Fourth ACM International Workshop on UnderWater Networks,* ser. WUWNet '09, no. 1. New York, NY, USA: ACM, 2009, pp. 1–8.

[16] E. Jacobsen, "On local interpolation of DFT outputs, available online: http://www.ericjacobsen.org/ftinterp.pdf," 2002.

[17] E. Jacobsen and P. Kootsookos, "Fast, accurate frequency estimators," *IEEE Signal Processing Magazine*, vol. 24, no. 3, pp. 123–125, May 2007.

[18] L. Fan and G. Qi, "Frequency estimator of sinusoid based on interpolation of three DFT spectral lines," *Signal Processing*, vol. 144, pp. 52–60, 2018.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on April 19,2026 at 09:17:59 UTC from IEEE Xplore. Restrictions apply.